<u>CliniScan</u>

Milestone 1 — Data Understanding & Preprocessing Pipeline

Project Title: CliniScan — Chest X-ray Abnormality Detection System

Dataset Used: VinBigData Chest X-ray Abnormalities Detection Dataset Hosted on Kaggle

Milestone Scope

✓ Dataset Understanding
✓ Annotation Analysis
✓ DICOM Handling
✓ Image Preprocessing Pipeline
✓ YOLO Annotation Conversion
✓ Code Implementation

PART A — DATASET UNDERSTANDING

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1. Background and Motivation

Chest radiography (CXR) is one of the most widely used diagnostic imaging techniques in modern medicine. It is essential for detecting pulmonary infections, structural abnormalities, cardiovascular enlargement, pleural diseases, and oncological findings.

However, interpreting chest X-rays requires expert radiologists and is time-intensive, especially in high-volume healthcare systems. Automated detection systems can assist clinicians by:

- Highlighting suspicious regions
- Reducing diagnostic workload
- Increasing consistency
- Supporting early detection

The VinBigData dataset was designed specifically to enable research in **automated abnormality localization** rather than simple image classification.

CliniScan builds its preprocessing foundation on this dataset.

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2. Dataset Composition

The dataset is structured into training and testing partitions.

Component	Description
Training Set	15,000 DICOM chest X-ray images
Test Set	3,000 DICOM images
Annotation File	train.csv
Annotation Style	Bounding Boxes
Radiologists	3 per image
Project Subset Used	50–60 images (development phase)

For Milestone-1, a controlled subset of 50–60 images was used to:

- Validate preprocessing pipeline
- Test annotation conversion
- Ensure computational efficiency

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3. Project Folder Structure (As Implemented)

```
b13-cliniscan/
|
├── data/
|  ├── raw/
| |  └── train/ (Kaggle DICOM images)
| |
|  ├── processed/
| |  └── png-images/
| |
| └── train.csv
|
├── docs/
|  ├── Dataset_Description.pdf
| └── Milestone1_Report.pdf
|
├── src/
```

```
|  ├── dicom_to_png.py
|  ├── preprocessing.py
| └── annotation_conversion.py
|
├── README.md
├── requirements.txt
└── LICENSE
```

This modular structure separates:

- Raw data
- Processed data
- Source code
- Documentation

This separation improves maintainability and reproducibility.

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4. Understanding DICOM Format

Images are stored in **DICOM (Digital Imaging and Communications in Medicine)** format — the global standard for medical imaging.

Unlike JPEG/PNG, DICOM files contain:

- Pixel data
- Metadata
- Acquisition parameters
- Imaging modality information

Important Technical Detail

The field PhotometricInterpretation can be:

- **MONOCHROME2** → Higher pixel value = Brighter
- **MONOCHROME1** → Higher pixel value = Darker (Inverted)

If inversion is not handled correctly, the lung fields appear reversed (white background, dark lungs), which can confuse a neural network.

During DICOM → PNG conversion, inversion is corrected when required.

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5. Annotation File — train.csv

The train.csv file contains bounding box annotations.

Each row represents:

- One abnormality
- Annotated by one radiologist
- On one image

Column Description

Column	Description
image_id	Unique identifier matching DICOM file
class_name	Abnormality name
class_id	Numeric label (0–14)
rad_id	Radiologist ID
x_min	Bounding box left coordinate
y_min	Bounding box top coordinate
x_max	Bounding box right coordinate
y_max	Bounding box bottom coordinate

Bounding boxes are defined in **absolute pixel coordinates** relative to the original image resolution.

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6. Pathology Classes

The dataset includes 14 thoracic abnormalities + 1 normal class.

ID	Class	Clinical Meaning
0	Aortic enlargement	Enlarged aortic silhouette
1	Atelectasis	Lung collapse
2	Calcification	Calcium deposits

ID	Class	Clinical Meaning
3	Cardiomegaly	Enlarged heart
4	Consolidation	Fluid-filled alveoli
5	ILD	Interstitial Lung Disease
6	Infiltration	Inflammatory opacity
7	Lung Opacity	Increased lung density
8	Nodule/Mass	Suspicious lesion
9	Other lesion	Misc abnormality
10	Pleural effusion	Fluid in pleural space
11	Pleural thickening	Thickened pleura
12	Pneumothorax	Collapsed lung
13	Pulmonary fibrosis	Lung scarring
14	No finding	Normal scan

Images labeled **"No finding"** contain no bounding boxes.

PART B — PREPROCESSING PIPELINE

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1. Importance of Medical Image Preprocessing

Raw medical images contain variability in:

- Resolution
- Intensity scaling
- Contrast
- Scanner artifacts
- Noise patterns

Deep learning models require:

- Uniform dimensions

- Consistent pixel distribution
- Reduced noise
- Preserved anatomical detail

Therefore, a structured preprocessing pipeline was implemented.

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2. Step-by-Step Pipeline

Step 1 — DICOM to PNG Conversion

Tool: pydicom + OpenCV

Purpose:

- Extract pixel array
- Correct inversion (if MONOCHROME1)
- Normalize intensity
- Save as 8-bit PNG

Why convert?

- PNG is lightweight
- Faster loading
- Compatible with OpenCV and PyTorch

Step 2 — Grayscale Loading

Although X-rays are grayscale, OpenCV loads images in 3-channel BGR by default.

We explicitly load as:

cv2.IMREAD_GRAYSCALE

This:

- Reduces memory
- Reduces computation
- Preserves correct format

Step 3 — Resizing to 512×512

Original resolutions vary (some > 3000×4000).

Why 512×512?

- Uniform batch training
- Computationally efficient
- Maintains sufficient anatomical detail

Interpolation used:

cv2.INTER_AREA

Best for downscaling.

Step 4 — Intensity Normalization (Min-Max)

Different scanners produce different pixel ranges.

Min-Max normalization ensures:

- Consistent brightness range
- Better convergence during training
- Stable gradient updates

Step 5 — CLAHE (Contrast Enhancement)

Chest X-rays often have low local contrast.

CLAHE improves:

- Vessel visibility
- Nodule detectability
- Lung texture clarity

Parameters:

- Clip limit: 2.0
- Tile grid: 8×8

This enhances subtle abnormalities without amplifying noise excessively.

Step 6 — Gaussian Blur (Denoising)

Medical imaging contains high-frequency quantum noise.

A light:

GaussianBlur (3×3)

removes noise while preserving edges.

Kernel size is small to avoid loss of lesion boundaries.

Step 7 — Z-Score Normalization

After enhancement and denoising, pixel values are standardized:

$$Z = \frac{X - \mu}{\sigma}$$

Benefits:

- Centers data distribution
- Improves neural network convergence
- Reduces training instability

Step 8 — Annotation Conversion to YOLO Format

Original bounding boxes are in corner format.

Converted to YOLO format:

class_id center_x center_y width height

Formula:

$$center_x = (x_{min} + x_{max})/2$$
$$width = x_{max} - x_{min}$$

All values normalized between 0 and 1.

Images with "No finding" → empty .txt files.

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3. Why This Order Is Important

1. DICOM conversion first — raw data extraction
2. Resize before normalization — ensures consistent scale
3. Normalize before CLAHE — improves histogram quality
4. CLAHE before blur — enhance then smooth noise
5. Annotation scaling after resize — correct coordinate mapping

Changing order may distort bounding box alignment.

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4. Implementation Files

File	Purpose
dicom_to_png.py	Convert raw DICOM to PNG
preprocessing.py	Apply resizing, CLAHE, blur, normalization
annotation_conversion.py	Convert CSV bounding boxes to YOLO

All scripts are modular and reusable.

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5. Output of Milestone 1

After completing this milestone:

- All selected DICOM images converted to PNG
- Images standardized to 512×512
- Contrast enhanced
- Noise reduced
- Bounding boxes converted to YOLO format
- Clean dataset ready for object detection training

Conclusion

Milestone-1 successfully established a structured and clinically aware preprocessing pipeline for chest X-ray abnormality detection.

By:

- Handling DICOM-specific challenges
- Standardizing image dimensions
- Enhancing local contrast
- Converting annotations properly

the dataset is now fully prepared for model development in subsequent milestones.

This forms the technical backbone of the CliniScan detection system.